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Application of data fusion based on deep belief network in air quality monitoring

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Abstract

The successive development of social industrialization and modernization has caused a great many of environmental problems. Due to the imbalance between the environment and rapid development, the problem of urban air quality has become more and more prominent, so air quality monitoring has become particularly important. In air quality monitoring, unexpected situations such as damage to the monitoring equipment and the relocation of the station building will occur, resulting in the lack of monitoring data. For this lack of monitoring data, the missing values of the monitoring data are supplemented by the data fusion method using deep belief network (DBN). It can provide relevant researchers with reference data for further analysis and research on air quality. Compared with BP neural network for data fusion, DBN method is closer to the actual monitoring value.

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Keywords: Deep belief network; data fusion; BP neural network; missing data; air quality monitoring

1. Introduction

The successive development of social industrialization and modernization, a large amount of polluting gases are discharged into the atmosphere, and air pollution problem is getting worse¹. Severe air pollution will affect the everyday life of human being, and in addition, it also brings certain obstacles to the sustainable development of society. The problem of air quality has attracted great attention from all walks of life². Air quality monitoring is

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particularly important because of the serious air quality problems that need to be addressed. Air quality monitoring can reflect the air quality, so as to achieve effective control and governance of air pollution problems.

For air quality related research, the source of air quality data is a very important part, and the actual monitoring data of air quality can be obtained through each monitoring station. The monitoring data observed by the air quality monitoring station can effectively reflect the real situation of the air quality in relevant areas around the monitoring station. Because of the complexity of monitoring data and the huge amount of data, as well as the continuous exploration and research of data fusion methods, the method of data fusion is applied in the field of air quality research, which has attracted widespread attention.

The concept of data fusion has been first mentioned many times in the military field, the data information of multiple sensors is fused together to produce a series of connections between the data³. The data fusion technology is more accurate and valuable than the data obtained by a single sensor. Scientific decision-making and estimation can be realized by integrating monitoring data from various sources at air quality monitoring points for analysis, synthesis, control and use^{4,5}. With the diversification of air quality data types, deep learning can conduct feature selection and correlation analysis on these complex air data, and build a deep learning network, which can depict rich internal information of the data. The deep learning algorithm can fuse data from multiple sources, which can better reflect the relationship between data than the traditional data fusion method.

In the air quality monitoring, it is inevitable to encounter monitoring instrument damage, station relocation and other unexpected situations, resulting in the lack of monitoring data. For this kind of data missing situation, related researchers lack some reference in the analysis and further research of air quality. Therefore, this paper adopts a method for fusion of multi-source data using deep belief network (DBN), which is applied in air quality monitoring. In the presence of missing atmospheric monitoring data, DBN model is adopted to supplement the missing value. Compared with the method of fusing multi-source data with BP neural network, the supplementary value of the DBN model is closer to the actual monitoring value, which can provide reference data for relevant researchers in the analysis and research of air quality.

2. Deep belief network (DBN)

Figure 1 is the network structure of the deep belief network (DBN). The DBN model is mainly formed by superimposing multiple restricted Boltzmann machines (RBM) ^{6,7}. For supervised training, a layer of BP neural network is added to the last layer. The process of DBN model is mainly composed of two parts, namely the pre-training process and the fine-tuning process⁸. Firstly, the unsupervised learning process is adopted to conduct pre-training for RBM network. Secondly, layers of pre-training has been carried out, the BP neural network is adopted in the last layer of DBN model for fine-tuning.

2.1. Restricted Boltzmann machine (RBM)

Figure 2 is the network structure of the restricted Boltzmann machine (RBM). It is mainly composed of two layers of neurons. One is the visible layer, which is adopted to input data containing features, the other is the hidden layer, which is adopted for reconstruct features⁹. For RBM, the nodes of each layer are not connected, and the nodes between each layer are all connected to each other. Therefore, all the neurons form a bipartite graph.

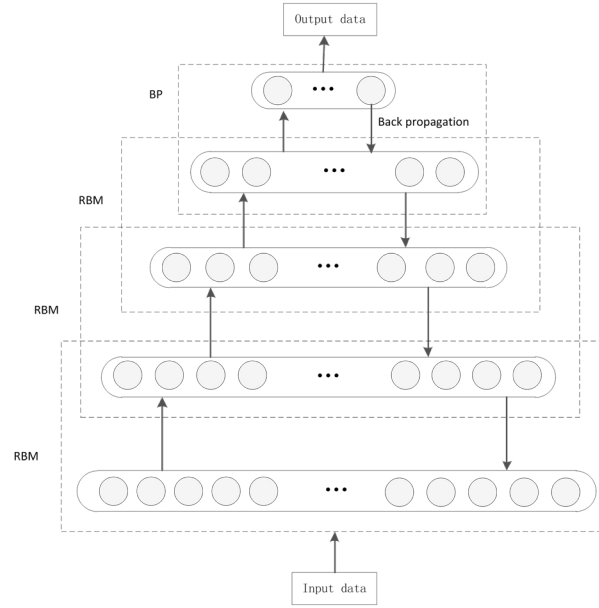


Fig. 1. DBN structure

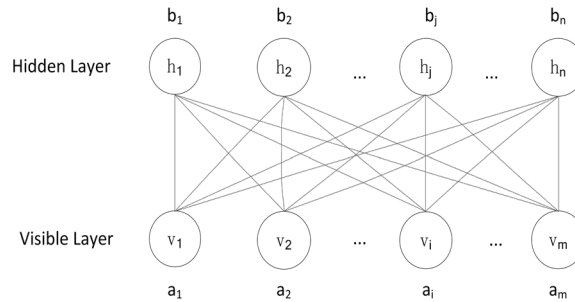


Fig. 2. RBM structure

The RBM model is an energy-based probability model. The training process is to treat each training sample as a state vector to maximize the probability of occurrence. Given a state (v, h) , suppose the number of neurons in the visible layer is m , suppose the number of neurons in the hidden layer is n . v_i is expressed as the state value of the i -th neuron node, h_j is expressed as the state value of the j -th neuron node⁹. $\theta = \{a_i, b_j, w_{ij}\}$ is expressed as various parameters in RBM, among them, in the visible layer, a_i is expressed as the bias of the i -th neuron node, and in the hidden layer, b_j is expressed as the bias of the j -th neuron node¹⁰, furthermore, w_{ij} is expressed as the weight of the connection between the i -th neuron node and the j -th neuron node. The energy function is expressed as:

$$E(v, h | \theta) = -\sum_{i=1}^m a_i v_i - \sum_{j=1}^n b_j h_j - \sum_{i=1}^m \sum_{j=1}^n h_j w_{ij} v_i \quad (1)$$

Through formula (1), the probability distribution of the joint can be obtained as follows:

$$P(v, h | \theta) = \frac{1}{Z(\theta)} e^{-E(v, h | \theta)} \quad (2)$$

$$Z(\theta) = \sum_{v, h} e^{-E(v, h | \theta)} \quad (3)$$

In formula (2), $Z(\theta)$ is expressed as an allocation function in RBM.

The probability of a single hidden layer unit being activated in RBM:

$$P(h_j = 1 | v) = \text{sigmoid}(b_j + \sum_i v_i w_{ij}) \quad (4)$$

The probability of a single visible layer unit being activated in RBM:

$$P(v_i = 1 | h) = \text{sigmoid}(a_i + \sum_j w_{ij} h_j) \quad (5)$$

2.2. BP neural network

BP neural network is used in many fields because of its strong adaptive ability, nonlinear mapping ability and generalization ability^{11,12}. In addition, it can adjust the error through backpropagation¹³.

The backpropagation of the BP neural network returns an error signal according to the original forward propagation path, and updates the weight and threshold of each neuron in each hidden layer, and the error signal tends to be minimized through the error function.

The error function is:

$$E = \frac{1}{2} \sum_{k=1}^N (Y_k - O_k)^2 \quad (6)$$

In formula (6), Y_k is expressed as the desired output, O_k is expressed as the actual output, and N is expressed as the number of neurons in the output layer.

3. Experiment and analysis

The monitoring data set is derived from meteorological data and atmospheric monitoring data in the environmental monitoring data of Shenyang, Liaoning Province. The meteorological data includes wind speed, humidity and temperature. The atmospheric monitoring data includes the concentration data of 6 pollutants: PM_{2.5}, PM₁₀, SO₂, CO, NO₂ and O₃.

The scope of monitoring data set is the hourly data per day from June 2019 to August 2019 and the hourly data per day from June 2020 to August 2020. Use part of the preprocessed monitoring data as training data and apply it in the training model, use another part of the preprocessed monitoring data as test data to evaluate the effect of the model.

3.1. Preprocessing data

In the analysis and use of monitoring data, the reliability of the training model and results can be ensured by eliminating the dimension of various data types. Therefore, normalization processing is required for atmospheric

monitoring data and meteorological data. A normalization method named Min-Max is used, which allows the data to be linearly transformed, and maps the used atmospheric monitoring data and meteorological data to (0,1). The formula for conversion is:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (7)$$

In formula (7), x_{norm} is expressed as the monitoring data after the normalization process, x is expressed as the original monitoring data sample. In the original monitoring data sample, x_{min} is expressed as the minimum value, and x_{max} is expressed as the maximum value.

3.2. Experimental process

The complete atmospheric monitoring data and meteorological data of Shenyang are normalized. In order to simulate the lack of one of the pollutant concentrations, the selected pollutant concentration in the complete data is used as a label. 80% of the processed data is randomly selected and input into the DBN model, and train a data fusion model based on DBN to supplement data for missing values. The experimental process is shown in Figure 3.

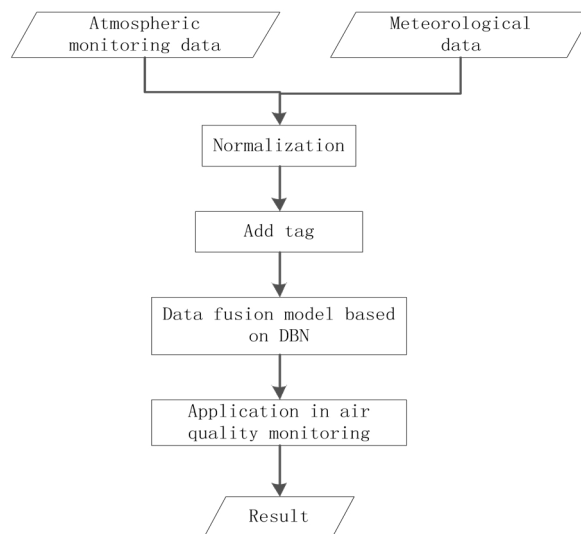


Fig. 3. Experimental flowchart

In the DBN model, the selected network node is 8-7-5-4-1. Among them, 8 input neurons and 1 output neuron. After verification, the effect of overlaying 3 layers of RBM is the best. The RBM network is pre-trained to initialize the weight of each layer node. The last layer of the DBN model is the BP neural network, which fine-tunes the model through backpropagation. In addition, the model uses Dropout to prevent overfitting. In RBM, the Sigmoid function is adopted as the activation function, which is applied to the visible layer and the hidden layer separately. And the formula is:

$$f(x) = \frac{1}{1 + e^{-x}} \quad (8)$$

Training is stopped until the training error is the minimum or the amount of training rounds is the maximum. Apply the data for testing to the trained model using DBN for multi-source data fusion. The output of this model is the value to supplement the missing pollutant concentration data. Compare the supplementary value of a certain data with the actual atmospheric monitoring value, and the model is evaluated through the comparison result.

3.3. Evaluation metrics

The effect of the model can be evaluated by the evaluation metrics. Two evaluation metrics, root mean square error (RMSE) and mean absolute error (MAE)¹⁴, are adopted in this paper. And the formulas are as follows:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_{bi} - y_i)^2} \quad (9)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |(y_{bi} - y_i)| \quad (10)$$

In formulas (9) and (10), N is expressed as the amount of experiments, y_{bi} is expressed as the supplementary value, y_i is expressed as the actual monitoring value.

RMSE is the square root of the ratio of the sum of the squares of the deviation between the supplementary value and the actual monitored value to the amount of experiments. It can be adopted to describe the deviation between the supplementary value and the actual atmospheric monitoring value, so as to reflect the degree of stability of the model.

MAE is the ratio of the sum of absolute errors of the supplementary value and the actual monitoring value to the amount of experiments. It can reflect the degree of quality of the model.

3.4. Result analysis

Use two different methods to supplement the missing values of atmospheric monitoring data, and the 6 different pollutants monitored are verified respectively. Table 1 is the results of experiments using two different models, and the results are evaluated by RMSE and MAE.

Table 1. The comparison results of the two models.

Model	Types of monitoring data	RMSE	MAE
BP neural network	PM _{2.5}	9.37	8.01
	PM ₁₀	9.81	8.15
	SO ₂	8.25	6.83
	CO	0.33	0.27
	NO ₂	11.45	10.13
	O ₃	12.05	10.66
DBN	PM _{2.5}	8.04	6.23
	PM ₁₀	8.44	6.58
	SO ₂	7.27	6.05
	CO	0.28	0.23
	NO ₂	9.05	7.83
	O ₃	9.78	8.03

Table 1 shows that, compared with the method using the BP neural network, the method of using the DBN model to fuse multi-source data to supplement the missing atmospheric monitoring data. The RMSE and MAE of the pollutant concentration have been reduced to varying degrees, indicating that the DBN model has relatively reduced the error.

4. Conclusion

Aiming at the lack of atmospheric monitoring data in air quality monitoring, this paper adopts a method of using the DBN model to fuse multi-source data to supplement the missing monitoring data. After experimental verification, it can be understood from the experimental comparison results that it is compared with the method of using BP neural network to fuse multi-source data, the method based on the DBN model has a smaller error with the actual monitoring value. Therefore, for further analysis and research on air quality, the DBN model is adopted to provide more effective reference data.

Acknowledgement

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